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## Oscillator with frequency control


### Abstract

A circuit includes a ring oscillator, a current source, an amplifier, a resistor-capacitor (RC) network, and a bias circuit. The current source is configured to power the ring oscillator. The amplifier is configured to control the current source. The amplifier has a first amplifier input and a second amplifier input. The RC network is coupled to the first amplifier input. The bias circuit is coupled to the second amplifier input. The bias circuit is configured to selectably set a frequency range of the ring oscillator, and selectably set a frequency of the ring oscillator within the frequency range.



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## Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims priority to U.S. Provisional Application No. 63/439,939, filed Jan. 19, 2023, entitled “Low Power Integrated Oscillator with Guaranteed Monotonic and Linear Frequency Control with Fine Step Size,” which is hereby incorporated by reference.

BACKGROUND

(1) Most digital circuits include an oscillator to provide clocking for synchronous operation. Different types of oscillators provide different levels of frequency accuracy. Crystal oscillators, and other oscillators using high-Q resonators, provide good frequency accuracy. Resistor-capacitor (RC) oscillators provide somewhat lower frequency accuracy, but offer other advantages. For example, a complete RC oscillator can be fabricated on an integrated circuit, which reduces application circuit complexity and cost.

SUMMARY

(2) In one example, a circuit includes a ring oscillator, a current source, an amplifier, a first capacitor, a first resistor, and a bias circuit. The ring oscillator has a power input. The current source has a control input and a current output. The current output is coupled to the power input. The amplifier has a first amplifier input, a second amplifier input and an amplifier output. The amplifier output is coupled to the control input of the current source. The first resistor and first capacitor are coupled in parallel to the first amplifier input. The bias circuit is coupled to the

second amplifier input. The bias circuit includes a second resistor, third resistor, fourth resistor, a fifth resistor, a first switch, and a second switch. The second, third, fourth, and fifth resistors are coupled in series. The first switch is coupled between the second amplifier input and a first terminal of the third resistor. The second switch is coupled between the second amplifier input and a second terminal of the third resistor.

(3) In another example, a circuit includes a ring oscillator, a current source, an amplifier, a resistor-capacitor (RC) network, and a bias circuit. The current source is configured to power the ring oscillator. The amplifier is configured to control the current source. The amplifier has a first amplifier input and a second amplifier input. The RC network is coupled to the first amplifier input. The bias circuit is coupled to the second amplifier input. The bias circuit is configured to selectably set a frequency range of the ring oscillator, and selectably set a frequency of the ring oscillator within the frequency range.

(4) In a further example, a microcontroller circuit includes a processor and an oscillator circuit. The oscillator circuit is coupled to the processor. The oscillator circuit includes a ring oscillator, a current source, an RC network, and a bias circuit. The current source is configured to power the ring oscillator. The amplifier is configured to control the current source. The amplifier has a first amplifier input and a second amplifier input. The RC network is coupled to the first amplifier input. The bias circuit is coupled to the second amplifier input. The bias circuit includes a coarse adjustment circuit and a fine adjustment circuit. The coarse adjustment circuit is configured to selectably set a frequency range of the ring oscillator. The fine adjustment circuit is configured to selectably set a frequency of the ring oscillator within the frequency range.

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## Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is a block diagram of an example oscillator with linear frequency control as described herein.

(2) FIG. **2** is schematic diagram of an example oscillator circuit with linear frequency control.

(3) FIG. **3** is a schematic diagram of an example alternative bias circuit suitable for use in the oscillator circuit of FIG. **2**.

(4) FIGS. **4** and **5** are graphs showing example coarse and fine frequency ranges in the oscillator circuit of FIG. **2**.

(5) FIG. **6** is a block diagram of an example microcontroller circuit that includes the oscillator circuit of FIG. **1**.

DETAILED DESCRIPTION

(6) To provide a stable frequency of oscillation, integrated resistor-capacitor (RC) oscillators base the oscillation frequency on resistance and capacitance values. The oscillation frequency is tuned by adjusting the resistance and/or the capacitance. However, the tuning may be nonlinear because oscillator frequency is proportional to RC **1** and the wider the tuning range, the greater the nonlinearity. Wide tuning range and fine step size (small frequency changes) are desirable in some applications. For example, fine step size allows low jitter in applications that modulate a fine trim code to produce a desired frequency. A linear tuning range may allow for equal jitter over the entire tuning range, and a large linear tuning range allows for greater toleration of process, voltage, and temperature variation. A large tuning range also supports the use of variable frequencies in a microcontroller or other digital circuit, where, for example, a higher frequency is used when more computation is required, and a lower frequency is used when less computation is required.

(7) The oscillator circuits described herein provide a wide tuning range with fine trim steps. The total tuning range of the oscillator circuits is divided into multiple smaller sub-ranges. Within the sub-ranges, the tuning steps may be small and monotonic.

(8) FIG. **1** is a block diagram of an example oscillator **100** with linear frequency control. The oscillator **100** includes a ring oscillator **102**, a current source **104**, an amplifier **106**, a resistor-capacitor (RC) network **108**, and a bias circuit **110**. The ring oscillator **102** generates an output signal (CLK) having a desired frequency. In some examples, the ring oscillator **102** includes an odd number of inverters connected in series, with an output of a last inverter in the ring (e.g., the inverter providing CLK) coupled to an input of the first inverter in the ring. The frequency of CLK is a function of the delay provided by the ring of inverters. The ring oscillator **102** may also be implemented using an even number of stages, where an odd number of the stages are inverting.
(9) The current source **104** provides a current that powers the ring oscillator **102**. A current output of the current source **104** is coupled to a power input of the ring oscillator **102**. A higher current increases the voltage across the ring oscillator **102**, which decreases the propagation delay of the ring oscillator **102**, and increases the frequency of CLK. A lower current decreases the voltage across the ring oscillator **102**, which increases the propagation delay of the ring oscillator **102**, and decreases the frequency of CLK. A control input of the current source **104** is coupled to the amplifier **106**.
(10) The amplifier **106** controls the current source **104** and the frequency of CLK. A first input of the amplifier **106** (e.g., the non-inverting input) is coupled to the RC network **108**. A second input of the amplifier **106** (e.g., the inverting input) is coupled to the bias circuit **110**. The RC network **108** is coupled to the ring oscillator **102**, and generates an output voltage based on the frequency of CLK. The bias circuit **110** generates a bias voltage based on a frequency selection signal (FREQ SEL). The bias circuit **110** may include a string of resistors, and FREQ SEL selects a tap point in the string of resistors from which the bias voltage is provided. The amplifier **106** generates an error signal representative of the difference of the RC network **108** output signal and the bias voltage provided by the bias circuit **110**. The error signal controls the current provided by the current source **104** to the ring oscillator **102**.
(11) The frequency of CLK is a function of the switched capacitance and resistance included in the RC network **108**, and the resistances provided in the bias circuit **110**. FREQ SEL selects the resistances in the bias circuit **110** to change the frequency of CLK. The frequency of CLK is linearly proportional to a resistance of the bias circuit **110** selected by FREQ SEL. The oscillator **100** can provide a wide total tuning range (e.g., 50%) with a small frequency step size (e.g., 0.1%), where the frequency steps are linear and monotonic.
(12) FIG. **2** is schematic diagram of an example oscillator circuit **200** with linear frequency control. The oscillator circuit **200** is an implementation of the oscillator **100**. The oscillator circuit **200** includes a ring oscillator **202**, a current source **204**, an amplifier **206**, an RC network **208**, a bias circuit **210**, and a control circuit **212**. The ring oscillator **202**, current source **204**, amplifier **206**, RC network **208**, and bias circuit **210** are examples of the ring oscillator **102**, current source **104**, amplifier **106**, RC network **108**, and bias circuit **110**, respectively. The ring oscillator **202** includes inverters **214**, **216**, and **218** coupled in series, with the output of each inverter coupled to the input of a next inverter, and the output of the inverter **218** coupled to the input of the inverter **214**. While the ring oscillator **202** is illustrated as including three inverters, various examples of the ring oscillator **202** may include any odd number of inverters.
(13) The current source **204** includes a transistor **220** coupled between a power terminal of the ring oscillator **202** and power terminal **266**. The transistor **220** may be a p-channel field effect transistor (PFET). Current flows through the transistor **220** to the ring oscillator **202** to produce the voltage powering the inverters **214**, **216**, and **218**. The frequency of CLK generated by the ring oscillator **202** is a function of the voltage powering the ring oscillator **202**.
(14) A control input (e.g., gate) of the transistor **220** is coupled to an output of the amplifier **206**. The amplifier **206** controls the current flowing through the transistor **220** to control frequency of CLK. A first input (e.g., non-inverting input) of the amplifier **206** is coupled to the RC network **208**, and a second input (e.g., inverting input) of the amplifier **206** is coupled to the bias circuit

**210**. The RC network **208** includes a resistor **222**, a capacitor **224**, a switch **226**, and a switch **228**. The resistance of the resistor **222** may be fixed (non-variable). The switch **226** and the switch **228** are coupled to the ring oscillator **202**, which provides intermediate signals as control signals ϕ1 and ϕ2, where ϕ1 is the inverse of ϕ2. ϕ1 controls opening and closing of the switch **226**, and ϕ2 controls opening and closing of the switch **228**. The oscillator circuit **200** may include level shifters that shift the control signals ϕ1 and ϕ2 to the power supply voltage VDD. The capacitor **224** is charged while the switch **226** is closed and discharged (through the resistor **222**) while the switch **228** is closed to develop the voltage provided at the first input of the amplifier **206**. The RC network **208** may also include the capacitor **264** in parallel with the resistor **222** to provide filtering.

(15) The bias circuit **210** provides a bias voltage to the amplifier **206**. The amplifier **206** generates an error signal representative of the difference in the bias voltage and output signal of the RC network **208**. The error signal controls the transistor **220**. The bias circuit **210** includes resistors coupled in series between the power terminal **266** and a ground terminal, and switches coupled to the resistors. The switches select the bias voltage provided to the amplifier **206**.

(16) The bias circuit **210** includes a coarse adjustment circuit **268** and a fine adjustment circuit **270**. The coarse adjustment circuit **268** includes resistors **230**, **232**, **234**, and **236**, and switches **246**, **248**, **250**, and **252**. The fine adjustment circuit **270** includes resistors **238**, **240**, **242**, and **244**, and switches **254**, **256**, **258**, and **260**. The resistors **230**, **232**, **234**, **236**, **238**, **240**, **242**, and **244** are coupled in series between the power terminal **266** and the ground terminal. The resistors **238**, **240**, **242**, and **244** may be unit resistors having a same value of resistance. The resistors **230**, **232**, **234**, and **236** may each have a same resistance that is greater than (e.g., an integer multiple of) the resistance of the unit resistor. For example, each of the resistors **230**, **232**, **234**, and **236** may include multiple (e.g., 2, 10, 100, etc.) unit resistors coupled in series. While the coarse adjustment circuit **268** and the fine adjustment circuit **270** are illustrated as including four resistors as a matter of convenience, various examples of the coarse adjustment circuit **268** and fine adjustment circuit **270** may include more resistors. For example, the fine adjustment circuit **270** may include 64, 100, or any other number of unit resistors coupled in series. The coarse adjustment circuit **268** may include 5, 10, or any other number of resistors coupled in series.

(17) The switch **246** is coupled to the node connecting the resistor **230** and the resistor **232** to form a first tap point. The switch **248** is coupled to the node connecting the resistor **232** and the resistor **234** to form a second tap point. The switch **250** is coupled to the node connecting the resistor **234** and the resistor **236** to form a third tap point. One of the switches **246**, **248**, **250**, or **252** is closed select a bias voltage provided to the amplifier **206**. The bias voltage may be expressed as: Vtap=X*VDD, where X is the division ratio selected by closing one of the switches **246**, **248**, **250**, or **252**.

(18) The switch **254** is coupled in parallel with the resistor **238**. The switch **256** is coupled in parallel with the resistor **240**. The switch **258** is coupled in parallel with the resistor **242**. The switch **260** is coupled in parallel with the resistor **244**. Accordingly, each of the switches **254**, **256**, **258**, or **260** may be closed to bypass the resistor in parallel with the switch. The switches **254**, **256**, **258**, or **260** are opened or closed to change the resistance between the ground terminal and the bias voltage tap point selected by closing one of the switches **246**, **248**, **250**, or **252**. For example, all of the switches of the fine adjustment circuit **270** are open to generate a first frequency of CLK, one of the switches of the fine adjustment circuit **270** is closed to generate a second frequency of CLK, two of the switches of the fine adjustment circuit **270** are closed to generate a third frequency of CLK, etc.

(19) The frequency of CLK generated by the oscillator circuit **200** (F.sub.CLK) may be expressed as:

(20) $F_{CLK} = \frac{R_2}{R_1 R_F C_F}$ (1) where: C.sub.F is the capacitance of the capacitor **224**; R.sub.F is the

resistance of the resistor **222**; R.sub.1 is the resistance between the power terminal **266** and the selected bias voltage tap point; and R.sub.2 is the resistance between the selected bias voltage tap point and ground.
(21) F.sub.CLK is linearly proportional to R.sub.2. A frequency range is selected to implement a coarse trim by closing one of the switches of the coarse adjustment circuit **268** (e.g., one of switches **246**, **248**, **250**, or **252**) to set R.sub.1 and an initial value of R.sub.2. Within the selected frequency range, fine frequency selection (fine trim) is provided by closing one or more of the switches of the fine adjustment circuit **270** (e.g., switches **254**, **256**, **258**, or **260**) to adjust R.sub.2.
(22) The control circuit **212** provides the control signals COARSE and FINE that control the switches of the coarse adjustment circuit **268** and the fine adjustment circuit **270**. The COARSE control signals may include as many signals as needed to control the switches of the coarse adjustment circuit **268**, and the FINE control signals may include as many signals as needed to control the switches of the fine adjustment circuit **270**. For example, the COARSE control signals may include a control signal for each switch of the coarse adjustment circuit **268**, and the FINE control signals may include a control signal for each switch of the fine adjustment circuit **270**. The control circuit **212** may be implemented in a microcontroller or other digital circuit. The control circuit **212** receives the signal FREQ SEL that specifies a value of F.sub.CLK, and the control circuit **212** sets the COARSE and FINE control signals to open/close the switches of the bias circuit **210** to select R.sub.1 and R.sub.2, and produce F.sub.CLK according to equation (1).
(23) FIG. **3** is a schematic diagram of an example alternative bias circuit **310** suitable for use in the oscillator circuit **200**. The bias circuit **310** is similar to the bias circuit **210** and may be used in the oscillator circuit **200** in place of the bias circuit **210**. In the bias circuit **310**, the coarse adjustment circuit **268** is the same in the bias circuit **210**. The bias circuit **310** includes the fine adjustment circuit **370** in place of the fine adjustment circuit **270** of the bias circuit **210**. The fine adjustment circuit **370** includes the resistors **238**, **240**, **242**, and **244**, and the switch **260** as in the bias circuit **210**. The switch **260** is coupled in parallel with the resistor **244** to bypass the resistor **244** when closed. The fine adjustment circuit **370** also includes the switches **354**, **356**, and **358**. The switch **358** is coupled in parallel with the resistors **244** and **242** to bypass the resistors **244** and **242** when closed. The switch **356** is coupled in parallel with the resistors **244**, **242**, and **240** to bypass the resistors **244**, **242**, and **240** when closed. The switch **354** is coupled in parallel with the resistors **244**, **242**, **240**, and **238** to bypass the resistors **244**, **242**, **240**, and **238** when closed. The control circuit **212** generates the COARSE control signals to select a bias voltage tap, and generates the FINE control signals to open or close the switches **260**, **358**, **256**, or **354** to select R.sub.1 and R.sub.2, and produce a desired F.sub.CLK in the oscillator circuit **200**.
(24) FIG. **4** is a graph of frequencies generated in an example of the oscillator circuit **200**. The y-axis represents frequency in megahertz and the x-axis represents fine trim step values. In FIG. **4**, the graph **402** shows frequency generated by the oscillator circuit **200** given selection of a first bias voltage tap of the bias circuit **210** (e.g., switch **246** closed, switches **248**, **250**, and **252** open). The graph **404** shows frequency generated by the oscillator circuit **200** given selection of a second bias voltage tap of the bias circuit **210** (e.g., switch **248** closed, switches **246**, **250**, and **252** open). The graph **406** shows frequency generated by the oscillator circuit **200** given selection of a third bias voltage tap of the bias circuit **210** (e.g., switch **250** closed, switches **246**, **248**, and **252** open). The graph **408** shows frequency generated by the oscillator circuit **200** given selection of a fourth bias voltage tap of the bias circuit **210** (e.g., switch **252** closed, switches **246**, **248**, and **250** open).
(25) The frequency ranges shown in each of the graphs **402-408** are selected by the COARSE control signals. In FIG. **4**, the fine adjustment circuit **270** includes 64 unit resistors in series and 64 switches that provide 64 fine frequency trim steps. The change in frequency with each fine trip step is based on the resistance of the each of the unit resistors of the fine adjustment circuit **270**. Similarly, the frequency spacing of the graphs **402-408** (the change in frequency with selection of successive bias voltage taps) is a function of the resistance of the resistors of the coarse adjustment

circuit **268**. In the implementation illustrated in FIG. **4**, the resistors of the coarse adjustment circuit **268** produce frequency ranges that are relatively close to one another with significant overlap.

(26) FIG. **5** is a graph of frequencies generated in another example of the oscillator circuit **200**. The y-axis represents frequency in megahertz and the x-axis represents fine trim step values. In FIG. **5**, the graph **502** shows frequency generated by the oscillator circuit **200** given selection of a first bias voltage tap of the bias circuit **210** (e.g., switch **246** closed, switches **248**, **250**, and **252** open). The graph **504** shows frequency generated by the oscillator circuit **200** given selection of a second bias voltage tap of the bias circuit **210** (e.g., switch **248** closed, switches **246**, **250**, and **252** open). The graph **506** shows frequency generated by the oscillator circuit **200** given selection of a third bias voltage tap of the bias circuit **210** (e.g., switch **250** closed, switches **246**, **248**, and **252** open). The graph **508** shows frequency generated by the oscillator circuit **200** given selection of a fourth bias voltage tap of the bias circuit **210** (e.g., switch **252** closed, switches **246**, **248**, and **250** open).

(27) The frequency ranges shown in each of the graphs **502-508** are selected by the COARSE control signals. In FIG. **5**, the fine adjustment circuit **270** includes 64 unit resistors in series and 64 switches that provide 64 fine frequency trim steps. The change in frequency with each fine trip step is based on the resistance of the each of the unit resistors of the fine adjustment circuit **270**. Similarly, the frequency spacing of the graphs **502-508** (the change in frequency with selection of successive bias voltage taps) is a function of the resistance of the resistors of the coarse adjustment circuit **268**. In the implementation illustrated in FIG. **5**, the resistors of the coarse adjustment circuit **268** produce frequency ranges that are spaced farther apart than the frequency ranges shown in FIG. **4**, with relatively little overlap between the frequency ranges.

(28) FIG. **6** is a block diagram of an example microcontroller circuit **600**. The microcontroller circuit **600** includes the oscillator **100** (or the oscillator circuit **200**), a processor **602**, and peripherals **604**. The processor **602** may be a general-purpose microprocessor core or any other instruction execution device or digital logic circuit. The peripherals **604** are coupled to the processor **602**. The peripherals **604** may be configured and controlled by the processor **602**, and the peripherals **604** may provide services to the processor **602**. For example, the peripherals **604** may include timers, communication circuits, interrupt control circuits, analog-to-digital converters, and other peripheral circuits.

(29) The oscillator **100** is coupled to the processor **602**. The oscillator **100** may also be coupled to the peripherals **604**. The oscillator **100** provides CLK to the processor **602**, and the processor **602** provides the signal FREQ SEL that the oscillator **100** uses to control adjustment of the frequency of CLK. The oscillator **100** provides low frequency variation over temperature with a wide tuning range and linear fine frequency trim.

(30) In this description, the term “couple” may cover connections, communications, or signal paths that enable a functional relationship consistent with this description. For example, if device A generates a signal to control device B to perform an action: (a) in a first example, device A is coupled to device B by direct connection; or (b) in a second example, device A is coupled to device B through intervening component C if intervening component C does not alter the functional relationship between device A and device B, such that device B is controlled by device A via the control signal generated by device A.

(31) Also, in this description, the recitation “based on” means “based at least in part on.” Therefore, if X is based on Y, then X may be a function of Y and any number of other factors.

(32) A device that is “configured to” perform a task or function may be configured (e.g., programmed and/or hardwired) at a time of manufacturing by a manufacturer to perform the function and/or may be configurable (or reconfigurable) by a user after manufacturing to perform the function and/or other additional or alternative functions. The configuring may be through firmware and/or software programming of the device, through a construction and/or layout of hardware components and interconnections of the device, or a combination thereof.

(33) As used herein, the terms “terminal,” “node,” “interconnection,” “pin,” and “lead” are used

interchangeably. Unless specifically stated to the contrary, these terms are generally used to mean an interconnection between or a terminus of a device element, a circuit element, an integrated circuit, a device or other electronics or semiconductor component.

(34) A circuit or device that is described herein as including certain components may instead be adapted to be coupled to those components to form the described circuitry or device. For example, a structure described as including one or more semiconductor elements (such as transistors), one or more passive elements (such as resistors, capacitors, and/or inductors), and/or one or more sources (such as voltage and/or current sources) may instead include only the semiconductor elements within a single physical device (e.g., a semiconductor die and/or integrated circuit (IC) package) and may be adapted to be coupled to at least some of the passive elements and/or the sources to form the described structure either at a time of manufacture or after a time of manufacture, for example, by an end-user and/or a third-party.

(35) While the use of particular transistors are described herein, other transistors (or equivalent devices) may be used instead with little or no change to the remaining circuitry. For example, a field effect transistor ("FET") (such as an n-channel FET (NFET) or a p-channel FET (PFET)), a bipolar junction transistor (BJT—e.g., NPN transistor or PNP transistor), insulated gate bipolar transistors (IGBTs), and/or junction field effect transistor (JFET) may be used in place of or in conjunction with the devices disclosed herein. The transistors may be depletion mode devices, drain-extended devices, enhancement mode devices, natural transistors, or other types of device structure transistors. Furthermore, the devices may be implemented in/over a silicon substrate (Si), a silicon carbide substrate (SiC), a gallium nitride substrate (GaN) or a gallium arsenide substrate (GaAs).

(36) References may be made in the claims to a transistor's control input and its current terminals. In the context of a FET, the control input is the gate, and the current terminals are the drain and source. In the context of a BJT, the control input is the base, and the current terminals are the collector and emitter.

(37) Circuits described herein are reconfigurable to include additional or different components to provide functionality at least partially similar to functionality available prior to the component replacement. Components shown as resistors, unless otherwise stated, are generally representative of any one or more elements coupled in series and/or parallel to provide an amount of impedance represented by the resistor shown. For example, a resistor or capacitor shown and described herein as a single component may instead be multiple resistors or capacitors, respectively, coupled in parallel between the same nodes. For example, a resistor or capacitor shown and described herein as a single component may instead be multiple resistors or capacitors, respectively, coupled in series between the same two nodes as the single resistor or capacitor.

(38) While certain elements of the described examples are included in an integrated circuit and other elements are external to the integrated circuit, in other example embodiments, additional or fewer features may be incorporated into the integrated circuit. In addition, some or all of the features illustrated as being external to the integrated circuit may be included in the integrated circuit and/or some features illustrated as being internal to the integrated circuit may be incorporated outside of the integrated. As used herein, the term "integrated circuit" means one or more circuits that are: (i) incorporated in/over a semiconductor substrate; (ii) incorporated in a single semiconductor package; (iii) incorporated into the same module; and/or (iv) incorporated in/on the same printed circuit board.

(39) Uses of the phrase "ground" in the foregoing description include a chassis ground, an Earth ground, a floating ground, a virtual ground, a digital ground, a common ground, and/or any other form of ground connection applicable to, or suitable for, the teachings of this description. In this description, unless otherwise stated, "about," "approximately" or "substantially" preceding a parameter means being within +/−10 percent of that parameter or, if the parameter is zero, a reasonable range of values around zero.

(40) Modifications are possible in the described embodiments, and other embodiments are possible, within the scope of the claims.

## Claims

1. A circuit comprising: a ring oscillator having a power input; a current source having a control input and a current output, the current output coupled to the power input, and an amplifier having a first amplifier input, a second amplifier input and an amplifier output, the amplifier output coupled to the control input of the current source; a first resistor and a first capacitor coupled in parallel and further coupled to the first amplifier input; and a bias circuit coupled to the second amplifier input, the bias circuit including: second, third, fourth, and fifth resistors coupled in series; a first switch coupled between the second amplifier input and a first terminal of the third resistor; a second switch coupled between the second amplifier input and a second terminal of the third resistor; a third switch coupled in parallel with the fourth resistor; and a fourth switch coupled in parallel with the fifth resistor, wherein: a first resistance of the second resistor is equal to a second resistance of the third resistor; a third resistance of the fourth resistor is equal to a fourth resistance of the fifth resistor; and the first resistance is larger than the third resistance.
2. The circuit of claim 1, wherein the fourth switch is further coupled to a ground terminal.
3. The circuit of claim 1, wherein the second, third, fourth, and fifth resistors are coupled in series between a power terminal and a ground terminal.
4. The circuit of claim 1, wherein: the first resistance is at least twice the third resistance.
5. The circuit of claim 1, wherein the first resistor has a fixed resistance.
6. The circuit of claim 1, further comprising: a fifth switch coupled between a first terminal of the first capacitor and a first terminal of the first resistor; and a sixth switch coupled between the first terminal of the first capacitor and a power terminal.
7. The circuit of claim 1, wherein the bias circuit is configured to: selectably set a frequency range of the ring oscillator by controlling the first switch and the second switch.
8. The circuit of claim 7, wherein the bias circuit is configured to: selectably set a frequency of the ring oscillator within the frequency range by controlling the third switch and the fourth switch.
9. A circuit comprising: a ring oscillator; a current source configured to power the ring oscillator; an amplifier configured to control the current source, the amplifier having a first amplifier input and a second amplifier input; a resistor-capacitor (RC) network coupled to the first amplifier input; and a bias circuit coupled to the second amplifier input and comprising: a first resistor, a second resistor, a third resistor, and a fourth resistor coupled in series; a first switch coupled between a first terminal of the second resistor and the second amplifier input; a second switch coupled between a second terminal of the second resistor and the second amplifier input; a third switch coupled in parallel with the third resistor; and a fourth switch coupled in parallel with the fourth resistor, wherein the bias circuit is configured to: selectably set a frequency range of the ring oscillator; and selectably set a frequency of the ring oscillator within the frequency range, and wherein: a first resistance of the first resistor is equal to a second resistance of the second resistor; a third resistance of the third resistor is equal to a fourth resistance of the fourth resistor; and the first resistance is larger than the third resistance.
10. The circuit of claim 9, wherein the bias circuit is configured to: close the first switch and open the second switch to selectably set the ring oscillator to a first frequency range; and close the second switch and open the first switch to selectably set the ring oscillator to a second frequency range.
11. The circuit of claim 8, wherein: the bias circuit is configured to: close the third switch to selectably set the frequency of the ring oscillator to a first frequency in the frequency range; and close the fourth switch to selectably set the frequency of the ring oscillator to a second frequency in the frequency range.

12. The circuit of claim 9, wherein: the fourth switch is further coupled to a ground terminal.
13. The circuit of claim 9, wherein: the first resistance is at least twice the third resistance.
14. The circuit of claim 9, wherein: the RC network includes: a fifth resistor coupled between the first amplifier input and a ground terminal; a capacitor coupled in parallel with the fifth resistor; a fifth switch coupled between the capacitor and a power terminal; and a sixth switch coupled between the fifth resistor and the capacitor.
15. The circuit of claim 9, wherein the first, second, third, and fourth resistors are coupled in series between a power terminal and a ground terminal.
16. A microcontroller circuit comprising: a processor; an oscillator circuit coupled to the processor, the oscillator circuit including: a ring oscillator; a current source configured to power the ring oscillator; an amplifier configured to control the current source, the amplifier having a first amplifier input and a second amplifier input; a resistor-capacitor (RC) network coupled to the first amplifier input; and a bias circuit coupled to the second amplifier input, the bias circuit including: a coarse adjustment circuit including: a first resistor and a second resistor coupled in series; a first switch coupled between a first terminal of the second resistor and the second amplifier input; and a second switch coupled between a second terminal of the second resistor and the second amplifier input; and a fine adjustment circuit including: a third resistor and a fourth resistor coupled in series with the first resistor and the second resistor; a third switch coupled in parallel with the third resistor; and a fourth switch coupled in parallel with the fourth resistor, wherein: a first resistance of the first resistor is equal to a second resistance of the second resistor; a third resistance of the third resistor is equal to a fourth resistance of the fourth resistor; and the first resistance is larger than the third resistance.
17. The microcontroller circuit of claim 16, wherein the first resistance is at least twice the third resistance.
18. The microcontroller circuit of claim 16, wherein the bias circuit is configured to: close the first switch and open the second switch to selectably set the ring oscillator to a first frequency range; and close the second switch and open the first switch to selectably set the ring oscillator to a second frequency range.
19. The microcontroller circuit of claim 16, wherein the coarse adjustment circuit is configured to selectably set a frequency range of the ring oscillator, and the fine adjustment circuit is configured to selectably set a frequency of the ring oscillator within the frequency range.
20. The microcontroller circuit of claim 19, wherein the bias circuit is configured to: close the third switch to selectably set the ring oscillator to a first frequency in the frequency range; and close the fourth switch to selectably set the ring oscillator to a second frequency in the frequency range.